Spring Framework 6

Beginner to Guru

Kubernetes Deployments

Kubernetes Pods

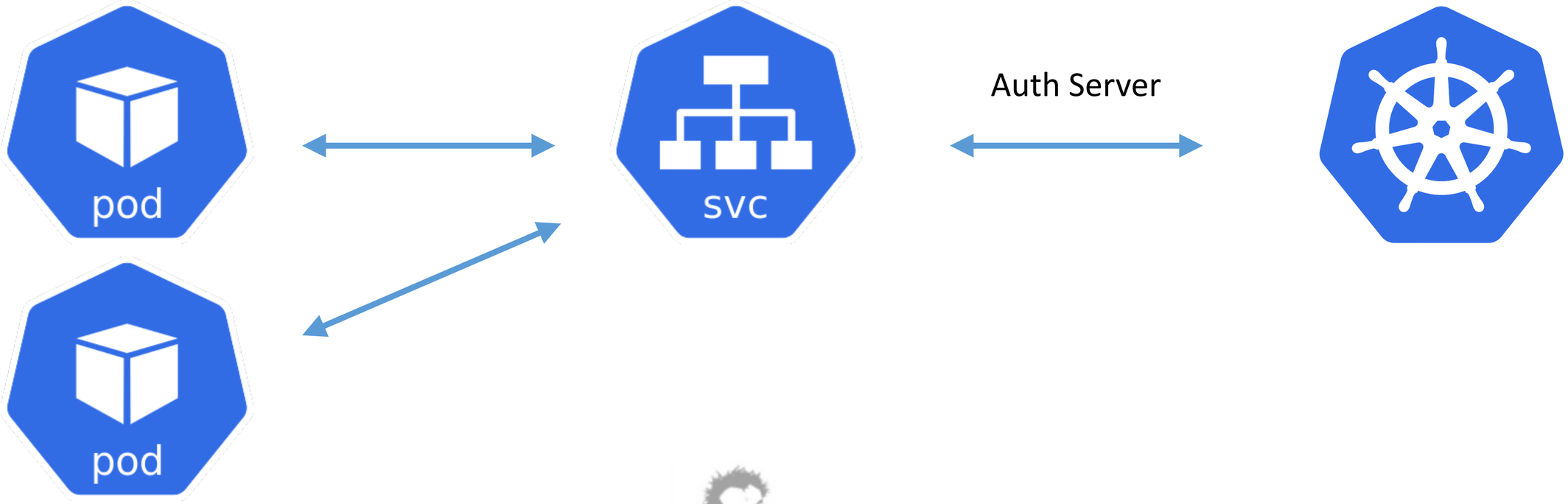
- **Pods:**
  - Groups of one or more containers
  - Most common is one container per pod
  - Can have 0 to N instances
  - Can run on one or more nodes (machines / VMs)

Kubernetes Services

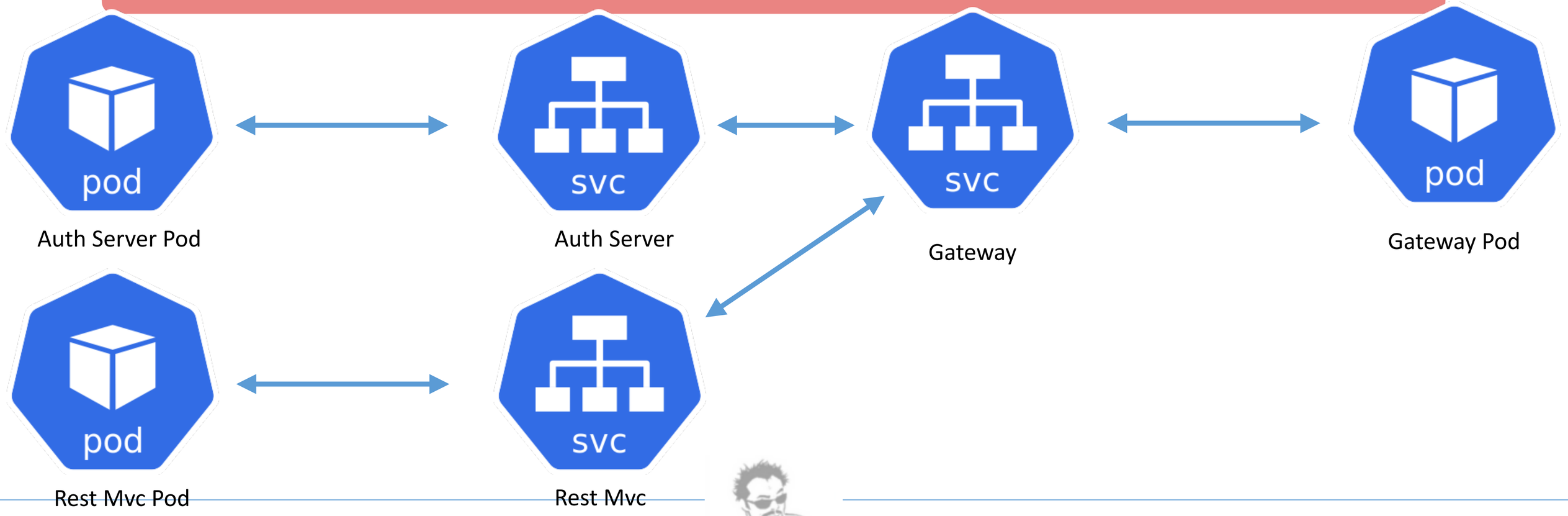
- **Services:**
  - A Service is a method for exposing to the network applications running in Pods
  - Pods get assigned an IP address, but are not permanent
  - Deployments, Node failures, scaling up / down will change Pod IPs
  - Services are an abstraction over Pods
  - Services also provide service discovery
    - ie An ‘auth-server’ service will be the host ‘auth-server’
    - Other services calling the host ‘auth-server’ will have their traffic routed to a pod instance

Single Pod Service

Multiple Pod Service



Gateway and Rest MVC



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Deploying to Kubernetes

- Pods and Services are deployed using the **`kubectl`** command
- Use **`kubectl`** to generate deployment yaml file
- Use **`kubectl`** to deploy deployment
- Use **`kubectl`** to create service definition file
- Use **`kubectl`** to deploy service

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